

Homework 1

Deadline: 11:59 pm September 6, 2026

Problem (1). *Guess a formula for*

$$1^2 + 3^2 + 5^2 + \cdots + (2n - 1)^2$$

(the sum of the squares of the first n odd numbers) by evaluating the sum for $n = 1, 2, 3, 4$. Then prove your formula by mathematical induction.

Problem (2). *Determine all positive integers n for which*

$$3^n > n^3.$$

Prove your claim by mathematical induction. Be careful: unlike the original exercise, the inequality is not monotonic in the small cases, so check $n = 1, 2, 3$ directly before deciding where an inductive argument can safely begin.

Problem (3). *Prove that $\sqrt{2} + \sqrt{3}$ is irrational.*

Hint: Suppose $\sqrt{2} + \sqrt{3} = r \in \mathbb{Q}$, isolate one of the radicals, and square.

Problem (4). *For $a, b \in \mathbb{R}$, prove that*

$$|a + b| = |a| + |b| \quad \text{if and only if} \quad ab \geq 0.$$

(That is, equality holds in the triangle inequality exactly when a and b have the same sign, or at least one of them is 0.)

Problem (5). *Prove by induction that for all integers $n \geq 2$ and all real numbers $a_1, a_2, \dots, a_n$,*

$$|a_1 - a_2 - a_3 - \cdots - a_n| \geq |a_1| - |a_2| - |a_3| - \cdots - |a_n|.$$

Hint: First establish the case $n = 2$, i.e., $|a - b| \geq |a| - |b|$ for all $a, b \in \mathbb{R}$, as a lemma from the ordinary triangle inequality; then use it in the inductive step.